

A Stability Theory of Chain-of-Thought Reasoning

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July 09, 2026

Abstract

Chain-of-Thought (CoT) prompting has emerged as a transformative technique for enhancing large language model (LLM) reasoning capabilities [Wei et al., 2022]. However, a fundamental theoretical question remains unresolved: when does adding reasoning steps improve accuracy versus amplify error? We introduce a stability theory for multi-step neural reasoning that formalizes this dichotomy. We model reasoning as a time-homogeneous Markov process operating on a state space of intermediate thoughts, and define the **Reasoning Stability Coefficient (RSC)**, a theoretical quantity that determines whether longer chains contract toward truth or expand into hallucination [Levin et al., 2017]. We prove three main results: (1) **exponential stability**, when $\text{RSC} < 1$ uniformly, deeper chains provably improve accuracy; (2) **error amplification** under a two-state coarse-graining where error persistence rate b exceeds escape rate a , error probability converges to stationary level $\pi_E = a/(a + (1 - b))$, approaching certainty in the near-absorbing regime where $1 - b \ll a$; and (3) **optimal depth** under an additional structural assumption combining contraction bias with linear noise accumulation, there exists a finite reasoning depth n^* beyond which additional steps degrade performance (“overthinking”) [Hassid et al., 2025]. We map our theoretical parameters to empirical LLM phenomena and derive practical implications for early stopping [Graves, 2016, Bajpai and Hanawal, 2025], self-consistency [Wang et al., 2023], and tool augmentation [Pronesti et al., 2026]. Our framework provides the first principled answer to when agents should reason longer versus stop.

1 Introduction

Large language models (LLMs) have demonstrated remarkable reasoning capabilities when prompted to generate intermediate steps, a technique known as Chain-of-Thought (CoT) prompting [Wei et al., 2022]. The intuition is straightforward: decomposing complex problems into sequential reasoning steps should improve accuracy by distributing computational load across the chain. Indeed,

empirical studies show that CoT prompting elicits reasoning in models that otherwise struggle with multi-step problems [Wei et al., 2022].

However, this intuition conceals a fundamental tension. While adding steps can improve reasoning, recent work reveals that longer chains exhibit “chain disloyalty,” abandoning correct reasoning paths [Lu et al., 2025] and suffer from “Late-Stage Fragility,” where later errors are more detrimental than early ones [Zhang et al., 2025]. Moreover, CoT prompting can obscure hallucination cues, creating a trade-off between reasoning depth and reliability [Cheng et al., 2025]. These observations suggest that longer chains do not always help; in some regimes, they actively harm performance by amplifying error [Hassid et al., 2025].

The core question we address is: *When should an agent reason longer versus stop?* Despite the empirical success of CoT, no theoretical framework exists to determine the stability properties of reasoning chains or to predict when additional steps will help versus hurt.

We introduce a **stability theory of Chain-of-Thought reasoning** that formalizes this dichotomy through three conceptual regimes:

1. **Stable regime:** Adding steps contracts expected distance-to-truth (improves accuracy).
2. **Unstable regime:** Adding steps amplifies error/hallucination (degrades accuracy).
3. **Knife-edge:** There exists an optimal depth n^* where marginal benefit equals marginal cost.

Our framework models reasoning as a **time-homogeneous Markov process** on a state space of intermediate thoughts. The kernel K_x depends on the problem x but crucially does **not** have access to the ground truth z , this reflects the epistemic limitation that models must reason without oracle access. We define the **Reasoning Stability Coefficient (RSC)** $\rho(x)$, a theoretical quantity measuring the expected contraction or expansion of distance-to-truth per reasoning step. When $\rho(x) < 1$ uniformly, the chain is a contraction mapping toward truth; when $\rho(x) > 1$, it expands away from truth.

We prove three main theorems: **(1)** Under uniform contraction ($\rho < 1$), expected error decreases exponentially with chain length; **(2)** Under a two-state coarse-graining where error persistence exceeds escape ($b > a$), error probability converges to stationary level $\pi_E = a/(a + (1 - b))$, approaching certainty when persistence is near-absorbing; and **(3)** Under an additional structural assumption combining geometric contraction with linear noise accumulation, there exists a finite optimal depth n^* that minimizes total error. This “overthinking” phenomenon, where additional reasoning degrades performance has been empirically observed [Hassid et al., 2025] but lacked theoretical explanation until now.

Our contributions are:

- A formal time-homogeneous Markov model of CoT reasoning with truth-coupled distance functionals
- The **Reasoning Stability Coefficient**, a theoretical quantity (approximable via verifiers) determining regime membership
- **Three theorems** characterizing stable, unstable, and optimal-depth regimes under explicit assumptions
- **LLM-specific implications** mapping theory to practice (early stopping, self-consistency, tool use)

2 Related Work

Chain-of-Thought and Extensions. Wei et al. [2022] introduced CoT prompting, showing that generating intermediate reasoning steps improves performance on complex tasks. Wang et al. [2023] extended this with self-consistency sampling multiple chains and aggregating via majority voting to improve robustness without additional training. Subsequent work generalized CoT to tree-structured reasoning [Tree-of-Thoughts Yao et al., 2023] and arbitrary graph structures [Graph-of-Thoughts Besta et al., 2024]. While these methods explore branching reasoning paths, our framework focuses on sequential chains, which remain the dominant inference pattern in deployed systems due to their simplicity and lower computational cost.

Error Accumulation in Reasoning. Recent empirical studies document failure modes in multi-step reasoning. Lu et al. [2025] identify “chain disloyalty” and metacognitive hallucinations that propagate through reasoning chains. Zhang et al. [2025] demonstrate “Late-Stage Fragility” errors in later steps are more detrimental than early errors, motivating adaptive correction strategies. Cheng et al. [2025] show that CoT prompting obscures hallucination cues, reducing detection effectiveness. Hassid et al. [2025] provide systematic evidence that shorter reasoning chains are more accurate than longer ones for the same questions, validating the “overthinking” phenomenon. Our theory provides the first formal explanation for these empirical observations.

Verification and Process Rewards. To mitigate error accumulation, recent work trains verifiers to check intermediate steps. Sharma et al. [2025] establish a PAC-learning framework for training CoT verifiers with sample complexity bounds. Pronesti et al. [2026] introduce Verifiable Process Reward Models (VPRMs) using deterministic rule-based verifiers, showing significant improvements over outcome-level rewards. Xie et al. [2024] use Monte Carlo Tree Search to collect step-level preference data, enabling fine-grained credit assignment. These approaches can be understood through our framework as mechanisms to reduce the error persistence rate b or estimate the distance-to-truth functional D_z .

Adaptive Computation. The idea of dynamic computation depth dates back to Graves [2016], who introduced Adaptive Computation Time (ACT)

for recurrent networks using “halting units” to optimize accuracy-computation trade-offs. Recent surveys [Bajpai and Hanawal, 2025] categorize early exit methods in deep learning, where intermediate classifiers enable adaptive inference. Our optimal depth theorem (Theorem 3) provides theoretical justification for these mechanisms in the specific context of stochastic reasoning chains.

Theoretical Foundations. Our analysis builds on classical Markov chain theory. The Doeblin-Dobrushin contraction coefficient [Levin et al., 2017] provides tools for analyzing convergence of Markov kernels, which we extend to non-stationary, truth-dependent settings. Recent work on f -divergence contraction [George, 2024] establishes bounds on information loss in Markov processes, supporting our definition of the Reasoning Stability Coefficient.

3 Formal Model

We formalize Chain-of-Thought reasoning as a **time-homogeneous Markov process** on a state space of intermediate thoughts. This abstraction is fully general: it captures token sequences, latent embeddings, or structured representations; it accommodates deterministic and stochastic transitions; and it subsumes special cases like self-consistency (via mixture distributions) and temperature sampling (via kernel smoothing).

3.1 State Space and Truth Representation

Let $\mathcal{S}$ denote the **reasoning state space**, comprising all possible intermediate representations (token sequences, latent thoughts, or structured summaries). Let $\mathcal{Z}$ denote the space of ground-truth solution objects (proofs, computations, correct answers). For each problem instance $x \in \mathcal{X}$, let $z = z(x) \in \mathcal{Z}$ denote the associated ground truth.

We define a measurable **answer mapping** $g : \mathcal{S} \rightarrow \mathcal{Y}$ that extracts the final answer $y = g(s_n)$ from terminal state s_n . The quality of a state s is measured by a **distance-to-truth functional** $D_z : \mathcal{S} \rightarrow \mathbb{R}_{\geq 0}$, where $D_z(s) = 0$ indicates perfect alignment with truth z .

Assumption 1 (Measurability and Boundedness). *The distance functional D_z is measurable with respect to the Borel σ -algebra on $\mathcal{S}$, and bounded: $\sup_{s \in \mathcal{S}} D_z(s) \leq D_{\max} < \infty$.*

Choice of D_z (Option A - Cleanest Theory). For this paper, we adopt:

$$D_z(s) = 1 - \Pr(\text{correct answer} \mid s) \quad (1)$$

This measures the probability that state s leads to an incorrect final answer. In practice, this can be estimated via learned verifiers [Sharma et al., 2025] or consistency checks. Alternative formulations (distance to proof manifolds, KL-divergence to target distributions) are discussed in Section 6.

3.2 Stochastic Reasoning Operator

Let $\mathcal{B}(\mathcal{S})$ denote the Borel σ -algebra on $\mathcal{S}$. A single reasoning step is modeled as a **time-homogeneous Markov kernel** $K_x : \mathcal{S} \times \mathcal{B}(\mathcal{S}) \rightarrow [0, 1]$:

$$K_x(s, A) = \Pr(s_{t+1} \in A \mid s_t = s, x) \quad (2)$$

Key Property: The kernel K_x depends on the problem x but **does not depend on the ground truth** z . This reflects the epistemic limitation that the model must generate reasoning without oracle access to z .

The kernel is fully general: it subsumes deterministic transitions (Dirac measures), autoregressive sampling (next-token distributions), and structured editing operations. The full reasoning chain is:

$$s_0 \sim \mu_x, \quad s_{t+1} \sim K_x(s_t, \cdot) \quad (3)$$

where μ_x is the initial state distribution (e.g., prompt encoding).

Assumption 2 (Time-Homogeneity). *The Markov kernel K_x is time-homogeneous: the transition probabilities do not depend on the step index t .*

3.3 The Reasoning Stability Coefficient (RSC)

We now define our central concept, the **Reasoning Stability Coefficient** $\rho(x)$ which quantifies whether reasoning steps contract toward or expand away from truth.

Definition 1 (Reasoning Stability Coefficient). For problem x with ground truth z , define:

$$\rho(x) = \sup_{s \in \mathcal{S} : D_z(s) > 0} \frac{\mathbb{E}[D_z(s_{t+1}) \mid s_t = s, x]}{D_z(s)} \quad (4)$$

Interpretation:

- $\rho(x) < 1$: **Contraction regime.** Each step reduces expected distance-to-truth. Longer chains improve accuracy.
- $\rho(x) > 1$: **Expansion regime.** Each step increases expected distance-to-truth. Longer chains amplify error.
- $\rho(x) = 1$: **Critical point.** Marginal steps neither help nor hurt in expectation.

Measurability: We assume the conditional expectation $\mathbb{E}[D_z(s_{t+1}) \mid s_t = s, x]$ is measurable in s and the supremum is finite (bounded by Assumption 1).

Important Note: The RSC is a **theoretical quantity** that depends on the unknown ground truth z . In practice, it must be approximated using learned verifiers, consistency checks, or held-out validation. We discuss estimation strategies in Section 6.

State-Dependent Extension. In practice, the contraction factor may vary across states. We can define a **local RSC**:

$$\rho(s) = \frac{\mathbb{E}[D_z(s_{t+1}) \mid s_t = s, x]}{D_z(s)} \quad (5)$$

The global RSC $\rho(x) = \sup_s \rho(s)$ represents the worst-case contraction. Uniform contraction (Assumption 3) is sufficient but not necessary for convergence; local contraction $\rho(s) < 1$ in reachable regions may suffice.

4 Main Results

We prove three theorems characterizing behavior in each regime. All proofs use standard tools from Markov process theory [Levin et al., 2017, George, 2024]; complete proofs are provided in Appendix A.

4.1 Theorem 1: Exponential Stability Under Uniform Contraction

Assumption 3 (Uniform Contraction). *There exists $\rho < 1$ such that for all $s \in \mathcal{S}$:*

$$\mathbb{E}[D_z(s_{t+1}) \mid s_t = s, x] \leq \rho \cdot D_z(s) \quad (6)$$

This is a strong global assumption requiring contraction at all states, not just in expectation over μ_x . In practice, $\rho(s)$ may vary; uniform contraction is sufficient but not necessary.

Theorem 1 (Exponential Stability). *Under Assumption 3 (uniform contraction with $\rho < 1$), for all $n \geq 0$:*

$$\mathbb{E}[D_z(s_n) \mid x] \leq \rho^n \cdot \mathbb{E}[D_z(s_0) \mid x] \quad (7)$$

Proof Sketch. By the tower property of conditional expectation and Assumption 3:

$$\mathbb{E}[D_z(s_{t+1})] = \mathbb{E}[\mathbb{E}[D_z(s_{t+1}) \mid s_t]] \leq \mathbb{E}[\rho \cdot D_z(s_t)] = \rho \cdot \mathbb{E}[D_z(s_t)] \quad (8)$$

Applying induction yields $\mathbb{E}[D_z(s_n)] \leq \rho^n \mathbb{E}[D_z(s_0)]$. $\square$

Meaning. Under uniform contraction, longer chains provably improve reasoning quality exponentially fast. The convergence is geometric with rate ρ , matching classical contraction mapping results [Levin et al., 2017]. This justifies “let’s think step by step” for tasks where $\rho < 1$ uniformly.

Realism Note. Uniform contraction is strong; it may hold locally in “easy” regions of $\mathcal{S}$ or for simple tasks. For hard reasoning tasks, $\rho(s)$ may vary across states, requiring localized analysis (see Section 6.2).

4.2 Theorem 2: Error Amplification

To analyze error amplification, we partition the state space into **correct region** $\mathcal{G} = \mathcal{S} \setminus \mathcal{E}$ and **error region** $\mathcal{E}$ (hallucinated, inconsistent, or invalid states).

Assumption 4 (Two-State Reduction). *Define the error indicator $e_t = \mathbf{1}_{\{s_t \in \mathcal{E}\}}$. Assume $\{e_t\}$ is a time-homogeneous Markov chain with constant transition probabilities:*

$$\Pr(e_{t+1} = 1 \mid e_t = 0) = a \quad (\text{escape rate}) \quad (9)$$

$$\Pr(e_{t+1} = 1 \mid e_t = 1) = b \quad (\text{persistence rate}) \quad (10)$$

This is an **explicit modeling assumption** and does not follow automatically from the Markov property of s_t ; it represents a coarse-graining approximation of the full process into correctness/error regions.

Theorem 2 (Error Accumulation Under Two-State Reduction). *Let $p_t = \Pr(e_t = 1)$. Under Assumption 4, the error probability evolves as:*

$$p_{t+1} = a + (b - a)p_t \quad (11)$$

with explicit solution:

$$p_n = \pi_E + (p_0 - \pi_E)(b - a)^n, \quad \text{where} \quad \pi_E = \frac{a}{a + (1 - b)} \quad (12)$$

If $|b - a| < 1$, then $p_n \rightarrow \pi_E$ geometrically with rate $|b - a|$.

Near-Absorbing Regime: If $b = 1 - \epsilon$ with $\epsilon \ll a$ (persistence is near-perfect, and leaving error is much rarer than entering it), then:

$$\pi_E = \frac{a}{a + \epsilon} \approx 1 - \frac{\epsilon}{a} \quad (13)$$

Thus, when error persistence is near-absorbing and entry into error at rate a dominates ϵ , the stationary error probability approaches certainty.

Proof. The recurrence $p_{t+1} = a + (b - a)p_t$ follows directly from law of total probability and Assumption 4. This affine recurrence has fixed point $p^* = a/(1 - (b - a)) = a/(a + (1 - b)) = \pi_E$. The solution is standard: $p_n = \pi_E + (p_0 - \pi_E)(b - a)^n$. Since $0 \leq a, b \leq 1$ and $(a, b) \notin \{(0, 1), (1, 0)\}$, we have $|b - a| < 1$, ensuring geometric convergence. $\square$

Lemma 1 (General One-Sided Bounds). *Without Assumption 4, define only:*

$$a = \sup_{s \in \mathcal{G}} \Pr(s_{t+1} \in \mathcal{E} \mid s_t = s) \quad (14)$$

$$b = \inf_{s \in \mathcal{E}} \Pr(s_{t+1} \in \mathcal{E} \mid s_t = s) \quad (15)$$

Then the error probability $p_t = \Pr(s_t \in \mathcal{E})$ satisfies:

$$bp_t \leq p_{t+1} \leq a + (1 - a)p_t \quad (16)$$

No closed-form solution or guaranteed monotonicity follows from these one-sided bounds alone.

Proof. Lower bound: $p_{t+1} \geq \Pr(s_t \in \mathcal{E}) \cdot \inf_{s \in \mathcal{E}} \Pr(s_{t+1} \in \mathcal{E} \mid s_t = s) = p_t \cdot b$.
Upper bound: $p_{t+1} \leq \Pr(s_t \in \mathcal{G}) \cdot a + \Pr(s_t \in \mathcal{E}) \cdot 1 = (1 - p_t)a + p_t = a + (1 - a)p_t$. $\square$

Interpretation. Theorem 2 provides exact dynamics under the two-state reduction, showing error probability converges to π_E . When $b > a$ and $\pi_E \approx 1$ (near-absorbing), this explains empirical “Late-Stage Fragility” [Zhang et al., 2025]. Lemma 1 provides honest guardrails: without the two-state assumption, we can only bound the dynamics, not characterize them fully.

4.3 Theorem 3: Optimal Depth and Overthinking

To formalize “more steps helps until it hurts,” we introduce an **additional structural assumption** decomposing total expected error into bias and variance terms.

Assumption 5 (Bias-Variance Decomposition). *Expected distance-to-truth decomposes as:*

$$\mathbb{E}[D_z(s_n)] \leq \rho^n D_0 + cn \quad (17)$$

where $D_0 := \mathbb{E}[D_z(s_0) \mid x]$ is the initial expected distance-to-truth, $0 < \rho < 1$ is the contraction rate, and $c > 0$ captures per-step irreducible noise (temperature sampling, model entropy, or perturbations).

The linear scaling assumes that per-step sampling errors are independent or weakly dependent across steps, yielding cumulative variance proportional to depth analogous to standard error growth in random walks. It is not derived from the Markov kernel alone but reflects an additional structural property of noisy reasoning systems.

Theorem 3 (Optimal Depth). *Under Assumption 5, the bound is minimized at finite depth:*

$$n^* = \left\lfloor \frac{\ln\left(\frac{D_0 |\ln \rho|}{c}\right)}{\ln(1/\rho)} \right\rfloor = \left\lfloor \frac{\ln\left(\frac{c}{D_0 |\ln \rho|}\right)}{\ln \rho} \right\rfloor \quad (18)$$

This formula applies when $D_0 |\ln \rho| > c$. If $D_0 |\ln \rho| \leq c$, the bound is minimized at $n^ = 0$ (no reasoning steps improve the bound).*

Proof Sketch. Consider the continuous relaxation $f(t) = \rho^t D_0 + ct$ for $t \in \mathbb{R}_{\geq 0}$. The derivative is:

$$f'(t) = \rho^t D_0 \ln \rho + c \quad (19)$$

Since $0 < \rho < 1$, we have $\ln \rho < 0$. Setting $f'(t^*) = 0$:

$$\rho^{t^*} D_0 |\ln \rho| = c \implies \rho^{t^*} = \frac{c}{D_0 |\ln \rho|} \quad (20)$$

Taking logarithms (using $\ln \rho = -\ln(1/\rho)$):

$$t^* = \frac{\ln\left(\frac{c}{D_0|\ln \rho|}\right)}{\ln \rho} = \frac{\ln\left(\frac{D_0|\ln \rho|}{c}\right)}{\ln(1/\rho)} \quad (21)$$

For integer n , check the discrete derivative:

$$\Delta f(n) = f(n+1) - f(n) = \rho^n D_0(\rho - 1) + c = -\rho^n D_0(1 - \rho) + c \quad (22)$$

The sign changes when $\rho^n D_0(1 - \rho) = c$. Using the exact expression for t^* and noting that $n^* = \lfloor t^* \rfloor$ or $\lceil t^* \rceil$ (whichever minimizes the discrete bound), we obtain the stated formula. The boundary condition $D_0|\ln \rho| > c$ ensures $t^* > 0$; otherwise $n^* = 0$. $\square$ $\square$

Meaning. There exists a provably optimal reasoning depth n^* for minimizing the bound in Assumption 5. Beyond this point, additional steps worsen the bound formalizing “overthinking.” This matches empirical findings that shorter chains often outperform longer ones [Hassid et al., 2025], and provides theoretical justification for early stopping mechanisms [Graves, 2016, Bajpai and Hanawal, 2025].

Approximation Note. For $\rho \approx 1$, we have $|\ln \rho| \approx 1 - \rho$, recovering the approximate formula $n^* \approx \ln(D_0(1 - \rho)/c)/\ln(1/\rho)$. However, the exact formula uses $|\ln \rho|$.

5 Implications for LLM Inference

We now map our theoretical parameters to empirical LLM phenomena and derive practical implications.

5.1 Parameter Interpretations

Escape rate a : Probability that a correct reasoning step introduces an error. Sources include:

- Sampling noise (temperature $T > 0$)
- Brittle arithmetic or symbolic manipulation
- Knowledge retrieval errors (hallucinated facts)
- Logical reasoning failures

Persistence rate b : Probability that an incorrect step remains incorrect. Mechanisms include:

- Self-reinforcement (model rationalizes previous errors)
- Confirmation bias in decoding (sampling tokens consistent with error)

- Autoregressive drift (error compounds in subsequent generation)

Stability coefficient $\rho(x)$: Task-dependent contraction/expansion determined by:

- Task difficulty (easy tasks contract, hard tasks expand)
- Model capability (stronger models have lower ρ)
- Prompt quality (well-structured prompts reduce ρ)

5.2 Early Stopping and Adaptive Depth

Theorem 3 implies that fixed-depth reasoning is suboptimal. Instead, agents should implement **adaptive depth** mechanisms [Graves, 2016, Bajpai and Hanawal, 2025]:

1. **Confidence-based stopping:** Halt when estimated $\hat{\rho}(s_t) \geq 1$ (stability deterioration)
2. **Verification gates:** Insert intermediate checks using learned verifiers [Sharma et al., 2025] or VPRMs [Pronesti et al., 2026] to detect when $s_t \in \mathcal{E}$
3. **Ponder cost optimization:** Trade accuracy against computation cost, as in Adaptive Computation Time [Graves, 2016]

The optimal depth formula provides a principled way to set maximum chain length based on estimated ρ and noise level c .

5.3 Self-Consistency and Ensemble Methods

Self-consistency sampling [Wang et al., 2023] reduces the effective escape rate a by marginalizing over multiple trajectories. In our framework:

- **Mechanism:** Sampling k independent chains and taking majority vote
- **Effect:** Under independence, majority voting reduces error probability exponentially in k , approximately $O(a^{\lceil k/2 \rceil})$ when $a < 1/2$. For small a , the error probability is dominated by the probability that at least $\lceil k/2 \rceil$ chains are in error.
- **Trade-off:** Increases computation cost; Theorem 3 suggests optimal k exists given cost constraints

For unstable tasks ($\rho > 1$), self-consistency provides limited benefit since all chains drift toward error. It is most effective in the knife-edge regime where $\rho \approx 1$.

5.4 Tool Augmentation and Verification

Tool use (calculators, search engines, code execution) affects stability parameters:

- **Reduces a :** External tools provide ground-truth for subproblems (arithmetic, fact lookup), reducing sampling-based errors
- **May increase b :** If tool outputs are unverified or incorrect, they can lock the chain into persistent error states (high b)
- **VPRMs [Pronesti et al., 2026]:** Deterministic rule-based verification can force $b \rightarrow 0$ by rejecting invalid states, shifting system to stable regime

Recommendation: Tools should be applied when estimated $\rho(x) > 1$ (unstable regime) to reduce a , but outputs must be verified to prevent increasing b .

5.5 Temperature and Sampling Strategy

Temperature T affects the noise parameter c in Assumption 5:

- **Low T (greedy):** Reduces c but may limit exploration, potentially increasing ρ if the model gets stuck in local optima
- **High T :** Increases c , reducing optimal depth n^* ; may help escape local optima but amplifies error in unstable regimes

For tasks with $\rho < 1$ (stable), higher temperature can be tolerated; for $\rho > 1$ (unstable), greedy decoding minimizes error amplification.

6 Limitations and Extensions

6.1 Estimating RSC in Practice

Our theory defines RSC as a theoretical quantity depending on the unknown ground truth z . In practice, we approximate it via:

1. **Learned verifiers:** Train classifiers to predict correctness from intermediate states [Sharma et al., 2025]
2. **Self-consistency checks:** Use agreement across multiple rollouts as proxy for confidence [Wang et al., 2023]
3. **VPRMs:** Apply deterministic rules to check step validity [Pronesti et al., 2026]

Future work should establish sample complexity bounds for RSC estimation from finite data, extending the PAC framework of Sharma et al. [2025].

6.2 Beyond Uniform Contraction

Assumption 3 (uniform contraction) is strong. Weaker alternatives include:

- **Local contraction:** $\rho(s) < 1$ only in region $\mathcal{S}_{\text{good}} \subset \mathcal{S}$
- **Expected contraction:** $\mathbb{E}_{s \sim \mu}[\rho(s)] < 1$ averaged over initial distribution
- **Lyapunov functions:** Find $V(s)$ such that $\mathbb{E}[V(s_{t+1})|s_t] \leq \rho V(s_t)$

These relaxations may yield weaker but more realistic convergence guarantees.

6.3 Beyond Sequential Chains

Our Markov model assumes sequential structure. Extensions to:

- **Tree-of-Thoughts [Yao et al., 2023]:** Requires branching process analysis; stability depends on exploration vs. exploitation trade-off
- **Graph-of-Thoughts [Besta et al., 2024]:** Convergence analysis on general graphs; may exhibit different stability properties due to aggregation effects
- **Revisable reasoning:** Allowing backtracking (MCTS [Xie et al., 2024]) changes the error model persistence rate b becomes controllable

6.4 Empirical Validation

While our framework explains existing empirical phenomena [Lu et al., 2025, Zhang et al., 2025, Cheng et al., 2025, Hassid et al., 2025], direct measurement of $\rho(x)$, a , and b for specific tasks remains future work. We propose:

- **Benchmarking:** Measure error rates vs. chain depth across task difficulty levels
- **Intervention studies:** Manipulate a (via tool use) and b (via verification) to test theoretical predictions
- **Model comparison:** Estimate RSC for different model families to characterize capability scaling

7 Conclusion

We have introduced the first stability theory for Chain-of-Thought reasoning, formalizing when longer chains help versus hurt. Our three theorems establish: (1) exponential improvement under uniform contraction, (2) error amplification under persistence, and (3) existence of optimal depth under bias-variance decomposition. The **Reasoning Stability Coefficient** $\rho(x)$ provides a theoretical

quantity (approximable via verifiers) for determining regime membership, with direct implications for inference strategies.

The “overthinking” phenomenon, where additional reasoning degrades performance is not merely an empirical curiosity but an inevitable consequence of noise accumulation in contractive systems [Hassid et al., 2025]. Our optimal depth formula (Theorem 3) provides a principled stopping criterion, while our parameter mapping (Section 5) translates theory into practice for early stopping [Graves, 2016, Bajpai and Hanawal, 2025], self-consistency [Wang et al., 2023], and tool use [Pronesti et al., 2026].

Future work should focus on empirical estimation of stability parameters and extension to non-sequential reasoning structures [Yao et al., 2023, Besta et al., 2024]. As LLMs are deployed in autonomous agents with extended reasoning horizons, understanding these stability properties becomes critical for reliable operation.

A Complete Proofs

A.1 Proof of Theorem 1

We prove by induction that $\mathbb{E}[D_z(s_n)] \leq \rho^n \mathbb{E}[D_z(s_0)]$.

Base case ($n = 0$): Trivially satisfied.

Inductive step: Assume $\mathbb{E}[D_z(s_t)] \leq \rho^t \mathbb{E}[D_z(s_0)]$. By the tower property and Assumption 3:

$$\mathbb{E}[D_z(s_{t+1})] = \mathbb{E}[\mathbb{E}[D_z(s_{t+1}) \mid s_t]] \leq \mathbb{E}[\rho \cdot D_z(s_t)] = \rho \cdot \mathbb{E}[D_z(s_t)] \leq \rho^{t+1} \mathbb{E}[D_z(s_0)] \quad (23)$$

where the first inequality uses Assumption 3 (uniform contraction) and the second uses the inductive hypothesis. $\square$

A.2 Proof of Theorem 2

Under Assumption 4, let $p_t = \Pr(e_t = 1)$. By law of total probability:

$$p_{t+1} = \Pr(e_{t+1} = 1 \mid e_t = 0) \Pr(e_t = 0) + \Pr(e_{t+1} = 1 \mid e_t = 1) \Pr(e_t = 1) \quad (24)$$

$$= a(1 - p_t) + bp_t = a + (b - a)p_t \quad (25)$$

This affine recurrence has fixed point:

$$p^* = \frac{a}{1 - (b - a)} = \frac{a}{a + (1 - b)} = \pi_E$$

The solution is:

$$p_n = \pi_E + (p_0 - \pi_E)(b - a)^n$$

Since $0 \leq a, b \leq 1$ and $(a, b) \notin \{(0, 1), (1, 0)\}$, we have $|b - a| < 1$, ensuring geometric convergence $p_n \rightarrow \pi_E$.

Near-Absorbing Regime: When $b = 1 - \epsilon$ with $\epsilon \ll a$:

$$\pi_E = \frac{a}{a + \epsilon} \approx 1 - \frac{\epsilon}{a}$$

Thus error probability approaches certainty when error persistence is near-perfect and leaving error is much rarer than entering it. $\square$

A.3 Proof of Theorem 3

Consider the continuous relaxation $f(t) = \rho^t D_0 + ct$ for $t \in \mathbb{R}_{\geq 0}$. The derivative is:

$$f'(t) = \rho^t D_0 \ln \rho + c \quad (26)$$

Since $0 < \rho < 1$, we have $\ln \rho < 0$. Setting $f'(t^*) = 0$:

$$\rho^{t^*} D_0 |\ln \rho| = c \implies \rho^{t^*} = \frac{c}{D_0 |\ln \rho|} \quad (27)$$

Taking logarithms (using $\ln \rho = -\ln(1/\rho)$):

$$t^* = \frac{\ln \left(\frac{c}{D_0 |\ln \rho|} \right)}{\ln \rho} = \frac{\ln \left(\frac{D_0 |\ln \rho|}{c} \right)}{\ln(1/\rho)} \quad (28)$$

For integer n , check the discrete derivative:

$$\Delta f(n) = f(n+1) - f(n) = \rho^n D_0 (\rho - 1) + c = -\rho^n D_0 (1 - \rho) + c \quad (29)$$

The sign changes when $\rho^n D_0 (1 - \rho) = c$. Using the exact expression for t^* and noting that $n^* = \lfloor t^* \rfloor$ or $\lceil t^* \rceil$ (whichever minimizes the discrete bound), we obtain the stated formula. The boundary condition $D_0 |\ln \rho| > c$ ensures $t^* > 0$; otherwise $n^* = 0$. $\square$

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